

Data Analytics Project

Author:	Catalina Herrera	Status:	Completed
Created:	25/06/2026	License:	Educational portfolio project
Tools:	SQL (DuckDB + DBeaver), Tableau Public, Git, GitHub		
Data Source:	Steam Games Dataset by FronkonGames (Kaggle)		

Project Overview & Version Control

In this fictional business scenario, Steam's Business Intelligence team aims to better understand the characteristics associated with commercially successful games available on the platform. By analyzing historical data related to pricing, genres, publishers, release dates, and player reception, the team aims to identify trends that can support strategic business decisions for both Steam and publishing partners.

This is an independent project created for portfolio purposes. Steam is not affiliated with or involved in this analysis. The business scenario, stakeholders, and recommendations presented in this document are fictional.

Data Source

This project uses the publicly available Steam Games Dataset published on Kaggle by FronkonGames. The dataset was created using the Steam Web API and Steam Spy, and includes published Steam games only (excluding DLCs, soundtracks, videos, and other

non-game content).

Version History

Version	Date	Description of Changes
v1.0	25/06/2026	Project planning, business questions definition and documentation setup
v2.0	26/06/2026	Data cleaning, SQL analysis, calculated metrics and business insights
v3.0	27/06/2026	Tableau dashboard creation, visualization refinement and final project documentation

Phase 1: Ask (Business Understanding)

1.1 Problem Statement

What is the core problem or challenge we are trying to solve?

Steam hosts thousands of games released every year, making it increasingly difficult for publishers and developers to identify the characteristics associated with commercial success and positive player reception. Understanding these factors can support data-driven pricing, publishing, and product development decisions.

What are the current organizational or technical pain points?

- Steam's marketplace has become increasingly competitive, making it difficult for publishers to understand which game characteristics influence commercial performance.
- Pricing strategies vary widely across genres.
- Publishers lack a consolidated view of how factors such as genre, price, release date, and player reviews relate to game success.

1.2 Business Objectives & Key Questions

Primary Objective: Identify the game characteristics most strongly associated with commercial success and player satisfaction to support pricing and publishing decisions.

Key Deliverables:

- SQL analysis
- Interactive Tableau dashboard
- Executive summary
- Business recommendations

Guiding Analytical Questions:

1. Which game genres are associated with the highest player ratings?
2. Is there a relationship between game price and player satisfaction?
3. Do free-to-play games receive better reviews than paid games?
4. Which publishers consistently release highly rated games?
5. How has the number of game releases evolved over time?

1.3 Stakeholders & Audience

- **Project Sponsor:** Steam Business Intelligence Team

- **Primary Stakeholders:** Steam Product Team, Steam Business Intelligence Team, Publishing Partners
- **End Users of Insights:** Business Intelligence Analysts, Product Managers, Marketing Team, Executive Leadership

Phase 2: Prepare (Data Sourcing & Collection)

2.1 Data Sources Inventory

Dataset Name	Source / System	Format	Size / Volume	Update Frequency
Steam Games Dataset	Kaggle (FronkonGames)	CSV	125.855 game records	Static

2.2 Data Credibility & Limitations

ROCCC Assessment: Is the data Reliable, Original, Comprehensive, Current, and Cited?

- **Reliable:** The dataset was compiled by FronkonGames using data collected from the Steam Web API and Steam Spy. Although it is not an official Steam dataset, it is based on publicly available data sources widely used for exploratory analysis.
- **Original:** The dataset is a compiled collection of publicly available Steam data rather than data published directly by Steam.
- **Comprehensive:** The dataset contains a wide range of variables, including pricing, genres, publishers, release dates, player reviews and ownership estimates, making it suitable for market analysis. The dataset includes published Steam games only. Downloadable content (DLCs), soundtracks, videos, episodes, and other non-game content were intentionally excluded.
- **Current:** The dataset represents a snapshot in time and may not include the most recent game releases or updates.
- **Cited:** The original dataset creator and Kaggle repository will be referenced throughout the project documentation.

Known Constraints: The analysis will be representative of a single snapshot and will not be updated in real time. Results should not be interpreted as official Steam business metrics. The dataset excludes DLCs, soundtracks, videos, and other non-game Steam content.

Data Governance & Security: The dataset contains no personally identifiable information. All data is publicly available and suitable for educational and portfolio purposes. No GDPR or privacy concerns were identified.

Phase 3: Process (Data Cleaning & Transformation)

3.1 Tools & Environment

- **Environment:** DuckDB, DBeaver, Tableau Public, Git, GitHub.
- **Key Packages/Libraries:** SQL, DuckDB, Tableau Public.

3.2 Cleaning Changelog & Execution

Step	Issue Identified	Action Taken / Transformation	Verification Method
1	CSV header contained a malformed column name (DiscountDLC count, missing a comma), causing column misalignment during import	Corrected the header by separating it into Discount and DLC count, then re-imported the dataset	Verified that all expected column names were imported correctly
2	Dataset contained 40 variables, many unrelated to the business objectives	Selected only the variables required to answer the analytical questions defined during the Ask phase	Verified resulting schema
3	The Reviews field contained non-null values for only 9.68% of records	Excluded from the cleaned dataset due to insufficient coverage and limited relevance to the analytical objectives	Verified non-null percentage using SQL aggregation
4	One record contained a missing game name (Name = NULL) and no	Excluded the record from the cleaned dataset	Verified that Name contains no NULL values in

Step	Issue Identified	Action Taken / Transformation	Verification Method
	meaningful analytical information		steam_games_clean
5	Checked for duplicate game records	Verified uniqueness using AppID. No duplicate AppID records were found	Compared total row count with distinct AppID count
6	Validated release date format	Converted Release date from text to date format and created release_year and release_month fields	Confirmed all 125,855 release dates were successfully converted
7	Validated numeric fields	Checked minimum and maximum values for Price, Discount, Peak CCU, and Average playtime forever	Confirmed values were within expected ranges: no negative prices, no discounts above 100%, and no negative playtime values
8	Created cleaned analysis dataset	Built steam_games_clean from selected variables, excluding one incomplete record and adding derived fields (total_reviews, positive_review_ratio, release_year, and release_month)	Verified final row count and reviewed resulting schema

Phase 4: Analyze (Exploratory Data Analysis)

4.1 Dataset Overview

Following the data preparation process, the cleaned dataset contains 125,854 Steam games and 21 analytical variables. The overview below summarizes the main characteristics of the dataset before addressing the business questions defined during the Ask phase.

For analysis involving player satisfaction, only games with at least 100 total reviews will be considered. This threshold helps reduce the influence of games with very few reviews, where the positive review ratio may not be representative.

Metric	Value
Total Games	125,854
Analytical Variables	21
Release Period	1997 - 2025
Free Games	26,660
Paid Games	99,194
Average Price	\$4.81
Average Positive Review Ratio	75.83%

4.2 Business Questions

Business Question 1: How has the number of game releases evolved over time?

The number of games released on Steam has grown substantially over time, reflecting the platform's rapid expansion. Between 1997 and 2005, only a handful of titles were released each year. From 2006 onwards, annual releases began to increase steadily, with particularly strong growth after 2014. Although there was a slight decline between 2018 and 2019, the overall upward trend continued. The most significant increase occurred between 2024 and 2025, when annual releases rose from 20,033 to 24,973 games.

The analysis also shows that 68.6% of all games in the cleaned dataset were released between 2021 and 2025, highlighting the rapid expansion of Steam's catalog in recent years. Due to the dataset being collected during 2026, that year was excluded from the trend analysis because it represents only a partial period.

Business Question 2: Which game genres are associated with the highest player ratings?

Although the dataset labels this field as "Genres", it also includes software categories, content descriptors, and Steam-specific classifications (e.g., Photo Editing, Movie, Free To Play). To answer the business question consistently, the analysis was restricted to the nine core video game genres.

- Casual has the highest average positive review ratio (81.02%), followed closely by Adventure (80.24%) and Indie (80.16%).
- The differences between genres are relatively small, with all nine core genres ranging from 77.13% to 81.02%. This suggests that player satisfaction is generally high across all major video game genres.
- Action is particularly interesting because, despite being one of the most represented genres (9,712 games), it has one of the lowest average review ratios (78.13%). This could indicate a greater variability in the quality of games published under that genre.
- Although Sports and Racing have comparatively lower review ratios, they also contain far fewer games (961 and 795, respectively), so their averages should be interpreted with more caution.
- Indie stands out for having the largest number of games (15,183) while also maintaining one of the lowest average numbers of reviews per game (3,742). This suggests that although Indie is a highly populated genre on Steam, player attention is distributed across many titles rather than concentrated on a few.

Business Question 3: Is there a relationship between price and player satisfaction?

A relationship between price and player satisfaction is observable, although it is not strictly

linear. Games priced between \$5.00 and \$9.99 achieve the highest average positive review ratio (82.20%), followed by games priced between \$10.00 and \$19.99 (81.57%). Satisfaction gradually decreases for higher price ranges, with games priced at \$60 or more receiving the lowest average rating (70.84%). Free-to-play games occupy a middle position, with an average positive review ratio of 77.18%.

These findings suggest that higher prices are not necessarily associated with higher player satisfaction. Instead, moderately priced games appear to receive the most favorable reviews, while premium-priced titles tend to achieve lower average ratings.

Note: The results for the \$40–59.99 and \$60+ price ranges should be interpreted with caution, as only 85 and 18 games, respectively, met the minimum review threshold of 100 reviews.

These results describe associations within the available dataset and should not be interpreted as evidence that price directly influences player satisfaction, as other factors, such as release year, may also affect review scores.

Business Question 4: Do free-to-play games receive better reviews than paid games?

Contrary to what might be expected, free-to-play games do not receive better player reviews than paid games. Among titles with at least 100 reviews, paid games achieve an average positive review ratio of 79.89%, compared with 77.18% for free-to-play games.

However, the difference between the two groups is relatively small (2.71 percentage points), suggesting that pricing alone is not a strong predictor of player satisfaction. Other factors, such as gameplay quality, genre, or player expectations, are likely to have a greater influence on review scores.

Business Question 5: Which publishers consistently release highly rated games?

To ensure an accurate publisher-level analysis, games with multiple publishers were split into individual publisher records before aggregation.

After filtering publishers with at least 10 games and considering only titles with 100 or more reviews, 27 publishers achieved an average positive rating above 90%.

To ensure an accurate publisher-level analysis, games with multiple publishers were split into individual publisher records before aggregation. After filtering publishers with at least 10 games and only titles with 100 or more reviews, 27 publishers achieved an average positive rating above 90%. Among them, NekoNyan Ltd., Adamvision Studios, and Mediascape Co., Ltd. obtained the highest average player ratings, all exceeding 95% positive reviews.

These results suggest that only a relatively small group of publishers consistently maintains exceptionally high player satisfaction across multiple releases.

Phase 5: Share (Data Visualization & Insights)

5.1 Primary Insights Summary

Insight / Core Observation	Supporting Evidence / Chart Reference
Steam's catalog has expanded rapidly over the last decade, with 67.7% of all games in the dataset released between 2021 and 2025	Chart 1 – Annual Game Releases
Casual games achieved the highest average player satisfaction (81.02%), although review scores remained relatively consistent across all major genres (77–81%)	Chart 2 – Average Player Rating by Genre
Higher prices were not associated with higher player satisfaction. Games priced between \$5 and \$9.99 received the highest average review scores, while premium-priced titles showed lower average ratings	Chart 3 – Player Satisfaction by Price Range
Games priced between \$5–\$19.99 achieved the highest average player satisfaction, while premium-priced games showed lower average ratings	Chart 3 – Player Satisfaction by Price Range
A small group of publishers consistently achieved outstanding player satisfaction, with 22 publishers maintaining an average positive review ratio above 90% while releasing at least 10 well-reviewed games	Chart 4 – Top Rated Publishers

5.2 Dashboard & Presentation Deliverables

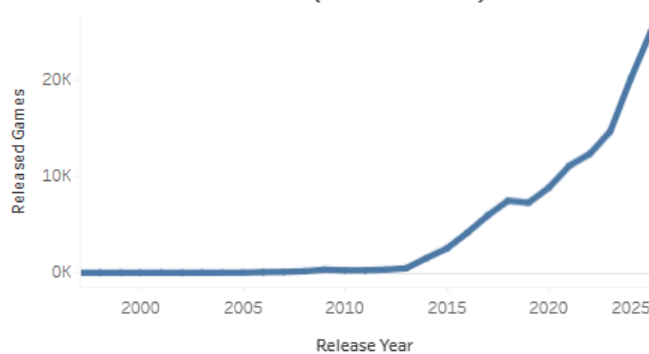
- **Interactive Dashboard Link:**
<https://public.tableau.com/app/profile/catalina.herrera1503/viz/SteamMarketTrendsPlayerSatisfaction19972025/Dashboard-SteamGamesAnalysis>

Steam Games Analysis

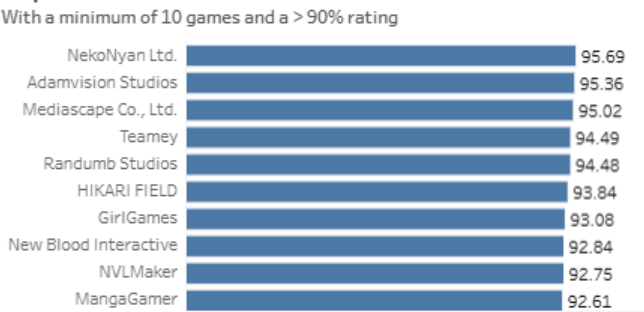
Steam Market Trends & Player Satisfaction (1997-2025)



Annual Game Releases (1997 - 2025)



Top Rated Publishers



Average Player Rating by Genre



Player Satisfaction by Price Range



Figure 1. Final Tableau dashboard summarizing the key business insights.

Phase 6: Act (Recommendations & Next Steps)

6.1 Strategic Recommendations

- 1. Prioritize quality over premium pricing:**
The analysis shows that higher-priced games do not necessarily receive better player ratings. Developers should focus on delivering high-quality gameplay and user experience rather than relying on premium pricing to increase perceived value. *(Based on: Business Question 3)*
- 2. Consider accessible pricing strategies:** Games priced between \$5 and \$9.99 achieved the highest average player satisfaction. Developers, particularly independent studios, may benefit from evaluating mid-range pricing strategies that balance affordability and perceived value. *(Based on: Business Question 3)*
- 3. Study successful Casual and Adventure titles:** Casual and Adventure games consistently achieved the highest average review scores among the core video game genres. Developers entering these genres should analyze successful titles to identify common design practices and player expectations. *(Based on: Business Question 2)*
- 4. Benchmark against high-performing publishers:** Publishers with consistently high player ratings can serve as valuable benchmarks for release strategy, quality assurance, and post-launch support. Studying their publishing patterns may help identify best practices associated with long-term player satisfaction. *(Based on: Business Question 5)*
- 5. Monitor long-term market growth:** The rapid increase in Steam releases since 2021 indicates a highly competitive marketplace. Developers should place greater emphasis on product differentiation, marketing, and community engagement to improve visibility among an expanding catalog of games. *(Based on: Business Question 1)*

6.2 Suggested Performance Indicators

- Average Positive Review Ratio (%)
- Average Price
- Games Released per Year
- Publisher Average Rating
- Peak Concurrent Users (Peak CCU)
- Recommendations